

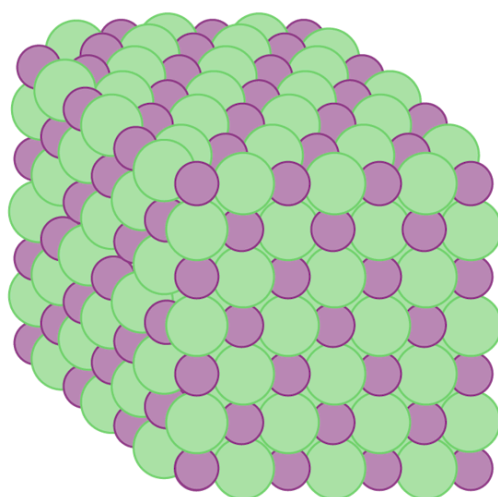
Covalent bonds

Learning outcomes

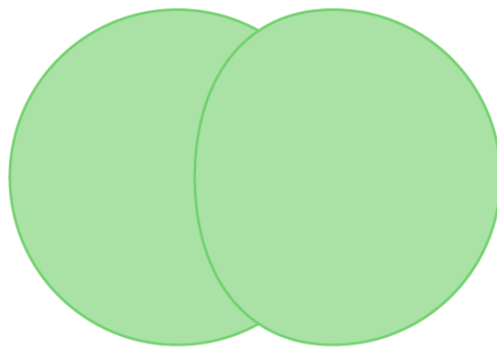
By the end of this section you should be able to:

- Define a covalent bond in terms of the electrostatic attraction and identify its role in allowing atoms to fulfil the octet rule by sharing valence electrons.
- Deduce the Lewis formula of molecules and ions for up to four electron pairs on each atom.
- Recognise and draw molecules which are stable despite having fewer than an octet of electrons.
- Determine the name of a simple molecular compound from its chemical formula.

In section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>), you learned how atoms could form an ionic bond when they gained or lost electrons to fulfil the octet rule. For example, an ionic bond forms between lithium and fluorine when lithium loses an electron and becomes a cation, while fluorine gains this electron and becomes an anion (**Figure 1**). These two oppositely charged ions then become attracted to each other in a crystal lattice, held by forces of electrostatic attraction.



 More information



 More information

Figure 1. Models of lithium fluoride crystal lattice and fluorine molecule.

Two fluorine atoms can also form a chemical bond with each other. Recall the electron configuration for a fluorine atom; it has seven valence electrons. You have learned before with ionic bonding that fluorine atoms easily gain an electron to complete the octet. However, if two fluorine atoms are bonding with each other, then how could both of them gain an electron to fulfil the octet rule? In this section, you will learn about how another type of chemical bond is formed instead.

Covalent bonding and the octet rule

When fluorine bonds to itself or to lithium, it has the same goal each time – fulfilling the octet rule. The octet rule states that elements will try to achieve a stable electron configuration by having 8 electrons in their valence shell, as noble gases do. This makes the element stable and less likely to react with other atoms.

You have already learned how elements can lose or gain electrons to achieve stability. However, fluorine cannot do this when it bonds with itself as each fluorine atom has 7 electrons in its outer shell. How could fluorine achieve this stability?

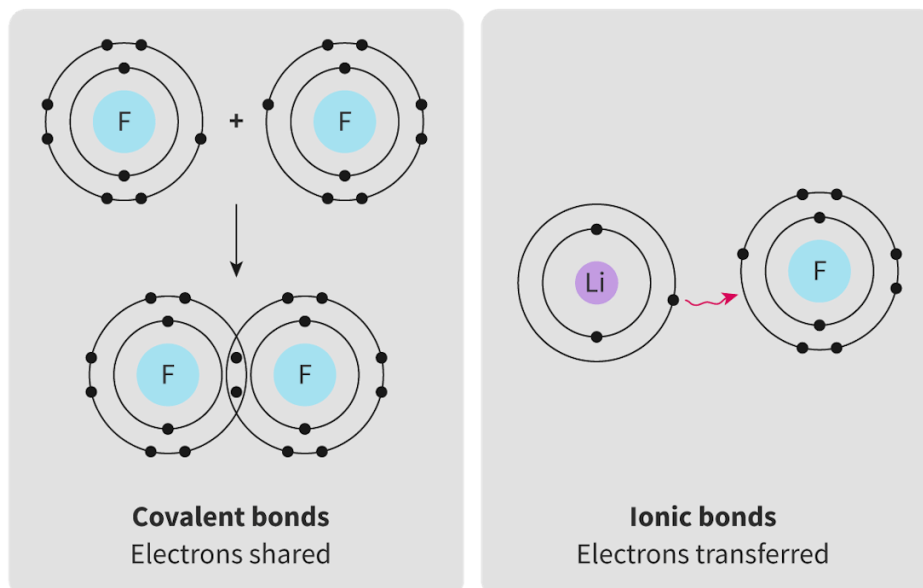


Figure 2. Ionic and covalent bonds.

📄 More information for figure 2

As you can see in **Figure 2**, fluorine will **share** electrons to achieve a stable octet. This electron sharing is a type of bond, known as a covalent bond and fluorine atoms contribute one electron each to a pair of electrons, which they then share.

Making connections

Use the ideas discussed above to explain why noble gases form covalent bonds less readily than other elements (see [section S1.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/>)).

Fluorine is not the only element which does this. For example, carbon can bond to other elements such as chlorine to form a molecule of CCl_4 (**Figure 3**), or to itself to form compounds such as diamond. Think about why this could be and read the **Making connections** box next.

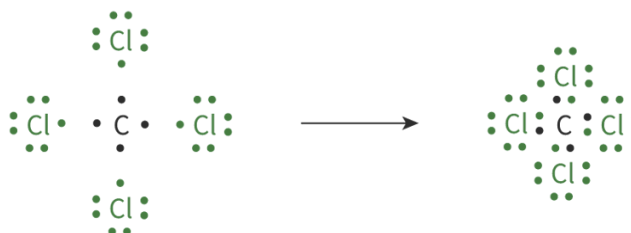


Figure 3. Use the image to identify how the CCl_4 molecule is formed using covalent bonds.

📄 More information for figure 3

Making connections

Why do ionic bonds only form between different elements while covalent bonds can form between atoms of the same element? (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)).

Now consider the electron configuration of a hydrogen atom. It has one electron occupying its first s orbital: $1s^1$. When two hydrogen atoms covalently bond, the s orbitals overlap (**Figure 4**) and the electron that each atom has contributed is now typically found in this shared space.

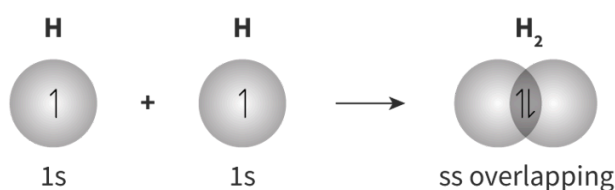


Figure 4. Covalent bonds involve the overlapping of orbitals. Each bond is made up of a pair of electrons, electron spin can be used to explain this.

📄 More information for figure 4

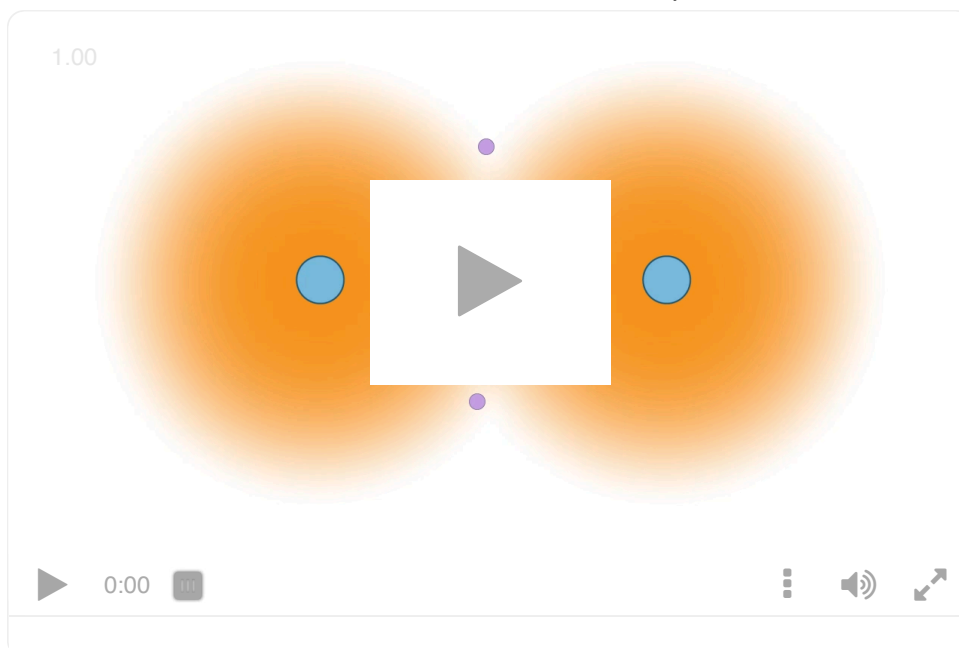
Note that when covalent bonds are being described it is always *pairs* of electrons, which are referred to as the bond.

Making connections

Think back to when you learned about electron configurations in [section S1.3.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>) and the importance of the direction of the electron spin from the Pauli exclusion principle. Try to use that concept to understand why stability is achieved when electrons are in pairs.

In an ionic bond it is the electrostatic attraction of cations and anions that holds two species together. View **Interactive 1** to see the electrostatic attraction of two positively charged nuclei.

In a covalent bond, the electrons are not gained or lost, so the atoms are neutral. How can it be that there would be electrostatic attraction between two neutral atoms?



Interactive 1. Electrostatic Attraction of Two Positively Charged Nuclei.

👁 More information for interactive 1

Recall from your study of [section S1.2.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>) that the nucleus contains protons, which are positively charged. When two electrons pair up to form a covalent bond, these negative electrons are found between the nuclei of those atoms and these are shared electrons.

Although molecules contain neutral atoms, there is still electrostatic attraction present in covalent bonds. In a covalent bond, it is the electrostatic attraction between the *positive* nuclei of two bonding atoms and their shared pair of *negatively* charged electrons that holds these atoms together.

Study skills

When asked to define any type of bonding you should always refer to what particles the electrostatic attraction is **between**. In an ionic bond this is between two oppositely charged ions. In a covalent bond it is between the positive nuclei of both atoms and their shared pair of electrons. Metallic bonding will be between a lattice of metal cations and a sea of delocalised electrons.

Lewis formula

As you will remember from [section S2.1.3a](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>) ionic compounds form giant crystal lattice structures that in theory can continue indefinitely.

Most compounds that are covalently bonded cannot do this. Instead, they will form smaller, discrete molecules. Much like an atom is the simplest unit with the fundamental chemical properties of an element, a molecule is the term used to describe atoms covalently bonded together.

At this level, most of the molecules you encounter will be composed of just a few atoms.

Concept

You may have noticed that the word 'molecule' has been used frequently in this subtopic but was not used in subtopic S2.1

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>).

The term molecule is only associated with elements and compounds which are bonded covalently. It represents a group of atoms which can exist as an independent entity. For example, the CO_2 molecules in fire extinguishers all exist as discrete molecules, i.e. that they exist separately from each other.

Ionic compounds are found in a crystal lattice structure. These lattices are found in varying sizes, and are represented using the term 'formula unit'. This is the simplest whole number ratio of cation to anion which exists. For example, NaCl is used to represent sodium chloride even though one grain of table salt contains many, many millions of ions joined together.

There are many ways to represent these molecules. One of the most widely used is the Lewis formula, also known as an electron dot or Lewis structure. This shows **all** valence electrons, whether these are pairs of electrons involved in covalent bonds or non-bonded pairs of electrons, known as non-bonding pairs.

Study skills

When drawing Lewis formulas make sure that you don't forget to include all non-bonded (lone) pairs of electrons. If you don't include them, then you are drawing a structural formula instead.

You also should not include any core electrons (electrons in inner shells) as this would be drawing a Bohr diagram (see **Figure 5**).

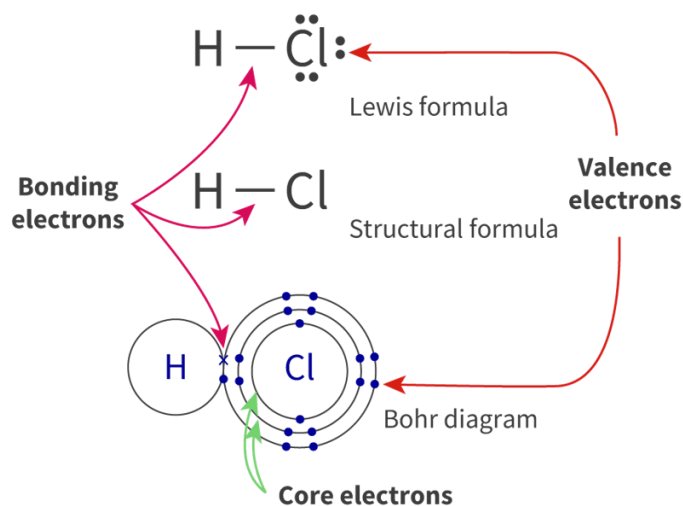


Figure 5. Different representations of hydrochloric acid: a Lewis formula, a structural formula and a Bohr diagram.

📖 More information for figure 5

In a Lewis formula, you can represent these electrons using dots, crosses, lines or a combination of these (**Figure 6**).

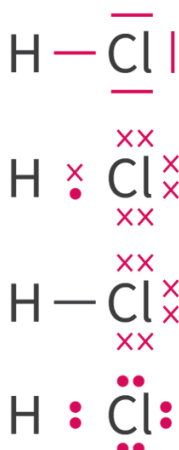


Figure 6. Different methods (dots, crosses, lines, combinations) of representing a molecule using Lewis formulas.

📖 More information for figure 6

Drawing Lewis formulas for binary molecules

To make sure that you always deduce the correct Lewis formula for a molecule composed of only two atoms, follow these steps:

1. Calculate the total number of valence electrons for each atom in the molecule (this is the number of valence electrons multiplied by the number of each atom).

2. Draw the skeletal structure of the molecule to show how the atoms are linked to each other with a single bond.
3. Add remaining pairs of electrons as non-bonding pairs to atoms that do not have a complete valence shell.
4. Check final structure contains correct number of valence electrons and all atoms have a filled valence shell. Remember, each bond represents two shared electrons.

Worked example 1

Draw the Lewis formula for hydrogen fluoride, HF.

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$H + F = 1 + 7$ $= 8 \text{ total}$
Step 2: Join all atoms with single bonds.	
Step 3: Add remaining electrons as non-bonding pairs (non-bonded electrons).	
Step 4: Check final structure is correct (e.g. contains 8 valence electrons total and valence shells for all atoms are complete).	$H = 1 \text{ bond} = 2 \text{ electrons}$ $F = 1 \text{ bond} + 3 \text{ non-bonding pairs} = 8 \text{ electrons}$

Molecules with more than two atoms

Molecules that are composed of three or more atoms will orient themselves such that one atom is in the middle, the 'central' atom and is surrounded by the other 'peripheral' atoms.

To determine the central atom:

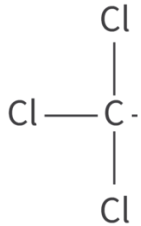
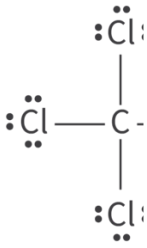
- The element that is furthest left on the periodic table is usually the central atom, e.g. the central atom in NO_2 is nitrogen as it is

further left than oxygen. Hydrogen is the one exception to this as it can never be a central atom.

- It will usually be the atom that has the lowest multiple, e.g. the central atom in H_2S is the sulfur.
- It may be the atom written first, e.g. the central atom in CH_4 will be the carbon.

Worked example 2

Draw the Lewis formula for tetrachloromethane, CCl_4 .

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{C} + (\text{Cl} \times 4) = 4 + (7 \times 4)$ $= 32 \text{ total}$
Step 2: Determine the central atom. Here it will be carbon as it has the lowest multiple, is furthest left on the periodic table and is written first in the formula. Join all peripheral atoms to the central carbon with single bonds.	
Step 3: Add remaining electrons as non-bonding pairs.	
Step 4: Check final structure is correct (e.g. contains 32 valence electrons total and valence shells for all atoms are complete).	$\text{C} = 4 \text{ bonds} = 8 \text{ electrons}$ each $\text{Cl} = 1 \text{ bond} + 3 \text{ non-bond}$

Worked example 3

Draw the Lewis diagram for hydrogen sulfide, H_2S .

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$(\text{H} \times 2) + \text{S} = (1 \times 2)$ $= 8 \text{ total}$
Step 2: Determine the central atom. Although it was not written first in the formula and is not furthest left on the periodic table, Sulfur is the central atom. This is because it has the lowest multiple and the alternative (hydrogen) can never be the central atom.	$\text{H} -$
Step 3: Join all atoms with single bonds.	$\text{H} -$
Step 4: Check final structure is correct (e.g. contains 32 valence electrons total and valence shells for all atoms are complete).	each $\text{H} = 1 \text{ bond} = 2 \text{ el}$ $\text{S} = 2 \text{ bonds} + 2 \text{ non-bonding} = 8 \text{ el}$

Sometimes, the octet rule is not followed...

Some molecules can achieve stability without following the octet rule. You encountered this with hydrogen and helium, which achieve stability with just two electrons in their outer shell.

Other atoms can also achieve stability with fewer than eight electrons in their outer shell. For example, beryllium is stable with four electrons in its outer shell and boron is stable with six outer electrons. Although these elements can form four single bonds, in certain molecules where they have incomplete octets, they are still stable.

Examples are beryllium dichloride, BeCl_2 , and boron trifluoride, BF_3 , shown in **Figure 7**. The boron atom has formed the maximum number of covalent bonds it possibly can when it forms three single bonds with three fluorine atoms. There is no other way for it to achieve the noble gas configuration. The BeCl_2 molecule also achieves stability with an incomplete octet.

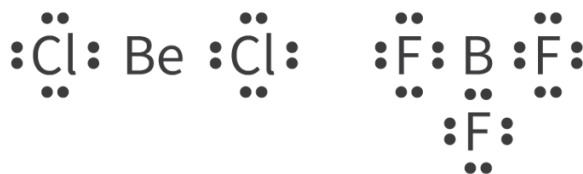


Figure 7. Lewis formulas of BF_3 and BeCl_2 .

🕒 More information for figure 7

Usually covalent bonds are formed between two elements that are non-metals, so why is beryllium bonding covalently when it is a metal?

In [section S2.2.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) you will learn more about how the level of attraction that nuclei have for a pair of electrons is not the same for all atoms, and how this determines the type of bond they will form. For now, think of Be as an exception that bonds covalently.

Aspect: Patterns and trends

Scientists analyse their observations, looking for patterns or trends, and try to draw general conclusions by inductive reasoning. They also look for discrepancies.

What are some of the limitations of the octet rule?

Drawing Lewis formula for polyatomic ions

When drawing the Lewis formula for polyatomic ions you should follow similar steps to those for simple molecules, but adjust the total number of valence electrons you are calculating to reflect the charge of the ion. Remember that cations have lost electrons and anions have gained electrons (**Figure 8**).



Figure 8. Eight valence electrons are seen in the Lewis formula for the hydroxide ion.

🕒 More information for figure 8

1. Calculate the total number of valence electrons in the ion (this is the number of valence electrons multiplied by the number of each atom).
2. Change the number of valence electrons to compensate for the charge (remember, each negative charge = add one electron; each positive charge = subtract one electron).
3. Determine the central atom (if applicable). Draw the skeletal structure of the molecule to show how the atoms are linked to each other. Join all atoms with single bonds.
4. Add remaining electrons (following the octet rule).
5. Add square brackets around the ion and the charge in the top right corner, outside the bracket.
6. Check final structure is correct (e.g. contains correct number of valence electrons total and valence shells for all atoms are complete).

Worked example 4

Draw the Lewis formula for the ammonium ion, NH_4^+ .

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{N} + (1 \times 4)$
Step 2: Adjust valence electrons to reflect charge of ion. (<i>Subtract one electron due to positive charge.</i>)	$9 - 1 = 8$
Step 3: Determine the central atom. Here it will be nitrogen as it has the lowest multiple, is written first in the formula and hydrogen cannot be a central atom. Join all atoms with single bonds.	

Solution steps	Calculations
Step 4: Add remaining electrons as non-bonding pairs (no change is needed here as all 8 electrons were used in step 3).	
Step 5: Add square brackets and charge.	
Step 6: Check final structure is correct (e.g. contains 8 valence electrons total and valence shells for all atoms are complete).	N = 4 l each H

Worked example 5

Draw the Lewis formula for the peroxide ion, O_2^{2-} .

Solution steps	Calculations
Step 1: Calculate total valence electrons	(O $\times$ 2) (6 $\times$ 2) = 12 total
Step 2: Adjust valence electrons to reflect charge of ion. (Add 2 electrons due to negative 2 charge.)	12 + 2 = 14
Step 3: Join all atoms with single bonds.	O :

Solution steps	Calculations
Step 4: Add remaining electrons as non-bonding pairs (2 electrons used to create single bond in step 3 so 12 electrons remaining out of total 14 electrons).	
Step 5: Add square brackets and charge.	
Step 6: Check final structure is correct (e.g. contains 14 valence electrons total and valence shells for all atoms are complete)	each O = 1 bond + 3 non-l

Making connections

Why do noble gases form covalent bonds less readily than other elements? Think about how bonding is related to the electron configurations you learned in [subtopic S1.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/).

Organic and inorganic compounds

Molecular compounds can be divided into organic and inorganic compounds (**Table 1**). Organic compounds are those that involve carbon and as part of [subtopic S3.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/) you will learn how to name these.

Table 1. Organic and inorganic compounds.

Organic compounds	Inorganic compounds
Must have carbon present and usually contains carbon bonded to hydrogen	May have carbon present but not bond hydrogen

You might be wondering about how molecular compounds are named. Here are some guidelines for naming inorganic molecular compounds.

Only one element

Some molecules are composed of only one type of element; all these molecules retain the name of the element. This is regardless of how many of each atom are present. For example, H_2 is referred to as hydrogen and Cl_2 is referred to as chlorine. A similar phenomenon occurs with two other molecules: phosphorus, P_4 , and sulfur, S_8 .

Study skills

A select group of elements always exist in pairs, which they are not paired with other elements. For example, oxygen will always exist as O_2 rather than as a single O, unless it's paired with another element such as in water, H_2O . These molecules are diatomic.

A useful mnemonic to remember which elements are diatomic is:

I Brought Chairs For Our New Home!

This stands for Iodine, Bromine, Chlorine, Fluorine, Oxygen, Nitrogen and Hydrogen, all of which exist as pairs (**Figure 9**).

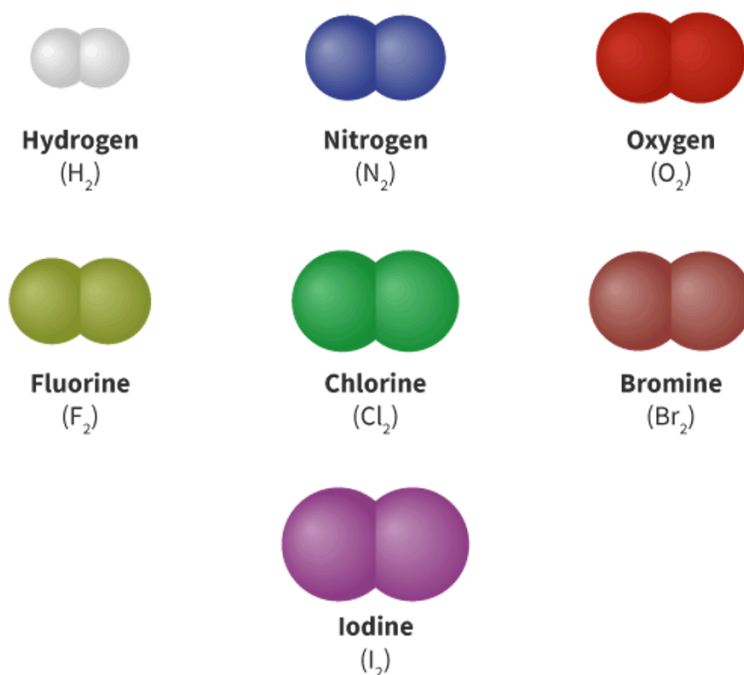


Figure 9. Elements that exist as diatomic molecules.

More than one element

When there is more than one present, we use an additional system involving prefixes to count the number of each atom present. For example, the molecule PCl_5 has one phosphorus and five chlorines present so the 'penta' prefix is added to the part of the name involving chlorine, phosphorus pentachloride. The exception is where there is only one atom for the first element in the molecule,

such as CO₂, it is called carbon dioxide, not monocarbon dioxide. Remember, if there is no prefix in front of an element, you should assume there is only one of that atom present (**Table 2**).

Table 2. Prefixes

Prefixes	
1	mono
2	di
3	tri
4	tetra
5	penta
6	hexa
7	hepta
8	octa
9	nona
10	deca

Common names

There are a number of compounds that are known by their common names. This includes water (H₂O), ammonia (NH₃) and ozone (O₃). As you can see, the names of the elements are not even included!

You might find it helpful to remember that in general, double vowels are not written, but there are some cases where they are, for example:

- ‘ao’ – drop the ‘a’ e.g. phosphorus pentoxide
- ‘ii’ – keep both ‘i’s e.g. nitrogen triiodide
- ‘oo’ – usually drop one ‘o’ e.g. nitrogen monoxide.

Worked example 1

State the name of the following compounds:

1. N_2
2. NO_2
3. N_2O
4. N_2O_3

1. N_2 = nitrogen
2. NO_2 = nitrogen dioxide
3. N_2O = dinitrogen monoxide
4. N_2O_3 = dinitrogen trioxide.

This next activity will allow you to practise drawing Lewis formulas for different models. Then, you can check your drawings by creating these models using the simulations.